

High Performance and Distributed Computing for Big Data

Unit 3: Cloud Systems and Big Data Management

Session 4 - AWS CLI + S3

Francesc Solsona Tehas francesc.solsona@udl.cat

Universitat Rovira i Virgili and Universitat de Lleida

- AWS S3: Storing files in the cloud
 - Installing AWS CLI
 - Configuring AWS credentials
 - Creating an S3 bucket
 - Syncing a local directory to upload to a bucket
 - Loading files from the bucket to the notebook from python
 - Writing files from the notebook to the bucket from python
 - Syncinc a local directory to download from a bucket

S3 - Storing files in the cloud

What is S3?

Amazon S3 (Simple Storage Service) is an object storage service that offers industry-leading scalability, data availability, security, and performance. It is designed to store and retrieve any amount of data from anywhere on the web.



S3 - Buckets and Objects

- **Buckets:** In Amazon S3, a bucket is a unique container for objects. Every object is stored within a bucket.
 - The bucket name must be globally unique across all existing bucket names in Amazon S3.
 - Buckets are used to store and organize objects.
- **Objects:** Objects are the fundamental entities within a bucket. They consist of data and metadata.
 - The information within an object is stored as a **key-value** pair.
 - The key, which is a unique identifier, is used to organize objects within the bucket. It is often formatted as a prefix to the object name.

For example, we can create a bucket called `hdc-b-{your-name}` and store objects organized by session or other criteria.

```
data-{your-name}/  
  project1/data.csv  
  project2/data.csv
```

Terms

- **Key** = *prefix + object name*
 - **prefix** = project1/ or project2/
 - **object name** = data.csv

Amazon S3 pricing is based on five factors:

1. **Storage Class:** The cost depends on the storage class used (Standard, Intelligent-Tiering, One Zone-IA, etc.).
2. **Storage:** The total volume of data stored per month.
3. **Requests:** The number and type of requests made.
4. **Data Transfer:** The cost of transferring data can vary by region and is also affected by whether data is transferred in or out.
5. **Management & Replication:** Additional features like data replication or management operations can also affect the cost.

For detailed information, you can refer to the [Amazon S3 Pricing Page](#).

AWS CLI

What is the AWS CLI?

The AWS CLI is a tool that allows you to interact with AWS services from the command line. It is a powerful tool that can be used to automate tasks and manage your AWS resources.

```
[ec2-user@ip-172-31-86-82 ~]$ aws
```

```
usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
```

To see help text, you can run:

```
aws help
```

```
aws <command> help
```

```
aws <command> <subcommand> help
```

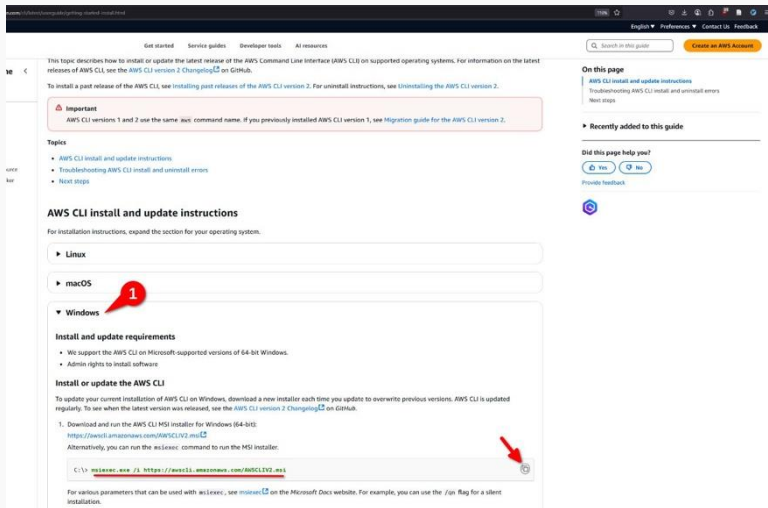
```
aws: error: the following arguments are required: command
```

```
[ec2-user@ip-172-31-86-82 ~]$
```

This is the base command which by itself doesn't do anything. What we are going to do now is to configure the AWS CLI with our credentials so we can later run commands that can access other AWS resources like S3.

Installing the AWS CLI

Visit the [AWS CLI installation guide](#) to install the AWS CLI on your machine.



The screenshot shows the AWS CLI installation guide page. The page is titled "AWS CLI install and update instructions". It includes a sidebar with navigation links and a main content area. The "Windows" section is expanded, showing "Install and update requirements" and "Install or update the AWS CLI". A red circle with the number "1" highlights the "Windows" section header. A red arrow points to the "msiexec" command in the "Install or update the AWS CLI" section.

Important
AWS CLI versions 1 and 2 use the same `aws` command name. If you previously installed AWS CLI version 1, see [Migration guide for the AWS CLI version 2](#).

Topics

- [AWS CLI install and update instructions](#)
- [Troubleshooting AWS CLI install and uninstall errors](#)
- [Next steps](#)

AWS CLI install and update instructions

For installation instructions, expand the section for your operating system.

- ▶ Linux
- ▶ macOS
- ▼ Windows

Install and update requirements

- We support the AWS CLI on Microsoft-supported versions of 64-bit Windows.
- Admin rights to install software

Install or update the AWS CLI

To update your current installation of AWS CLI on Windows, download a new installer each time you update to overwrite previous versions. AWS CLI is updated regularly. To see when the latest version was released, see [AWS CLI version 2 Changelog](#) on GitHub.

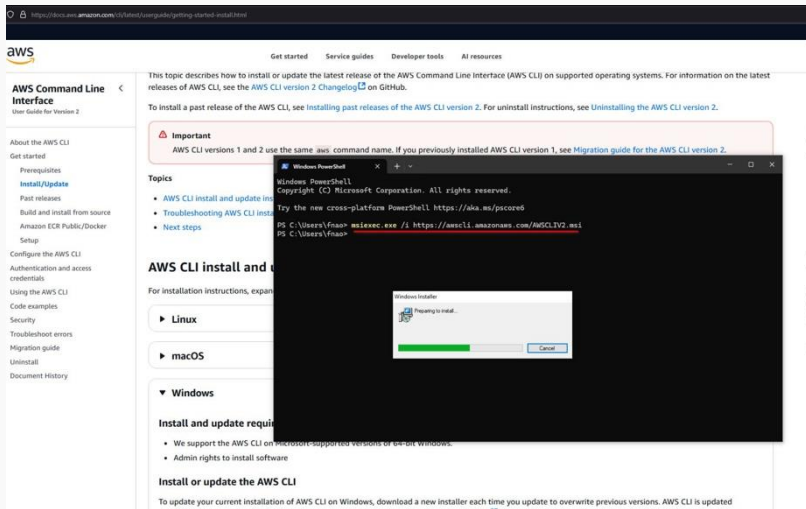
1. Download and run the AWS CLI MSI installer for Windows (64-bit):
<https://awscli.amazonaws.com/AWSCLIV2.msi>
Alternatively, you can run the `msiexec` command to run the MSI installer:

```
C:\> msiexec.exe /i https://awscli.amazonaws.com/AWSCLIV2.msi
```

For various parameters that can be used with `msiexec`, see [msiexec](#) on the Microsoft Docs website. For example, you can use the `/qn` flag for a silent installation.

Installing the AWS CLI

Paste the command on the terminal and wait for the installation to complete.



The screenshot shows the AWS Command Line Interface (CLI) user guide for Version 2 on the AWS website. The page is titled "AWS Command Line Interface" and includes a sidebar with navigation links such as "About the AWS CLI", "Get started", "Prerequisites", "Install/Update", "Past releases", "Build and install from source", "Amazon ECR Public/Docker", "Setup", "Configure the AWS CLI", "Authentication and access credentials", "Using the AWS CLI", "Code examples", "Security", "Troubleshoot errors", "Migration guide", "Uninstall", and "Document History".

The main content area of the guide includes an "Important" note stating that AWS CLI versions 1 and 2 use the same `aws` command name. It also provides links to the "AWS CLI version 2 Changelog" and "Installing past releases of the AWS CLI version 2".

Overlaid on the right side of the guide is a Windows PowerShell terminal window. The terminal shows the command to download the AWS CLI installer:

```
PS C:\Users\fnaao> msiexec.exe /i https://awscli.amazonaws.com/AWSCLIV2.msi
```

Below the terminal window, a "Windows Installer" dialog box is visible, showing the progress of the installation with a green progress bar and a "Cancel" button.

Installing the AWS CLI

Or if you are on MacOS, go to the MacOS section and also paste the corresponding commands on your terminal.

The screenshot shows the AWS Command Line Interface User Guide for Version 2. The left sidebar contains a navigation menu with links like 'About the AWS CLI', 'Get started', 'Prerequisites', 'Install/Update', 'Past releases', 'Build and install from source', 'Amazon ECR Public/Docker', 'Setup', 'Configure the AWS CLI', 'Authentication and access credentials', 'Using the AWS CLI', 'Code examples', 'Security', 'Troubleshoot errors', 'Migration guide', 'Uninstall', and 'Document History'. The main content area is titled 'macOS' and includes a section 'Install and update requirements' with two bullet points. Below this is a 'macOS version support matrix' table. Further down is a section 'Install or update the AWS CLI' with a paragraph and a code block containing two terminal commands. A red callout '1' points to the 'macOS' section header, and a red callout '2' points to the 'Command line installer - All users' tab. A red arrow points to a copy icon next to the terminal commands.

macOS

Install and update requirements

- We support the AWS CLI on macOS versions 11 and later. For more information, see [macOS support policy updates for the AWS CLI v2](#) on the AWS Developer Tools Blog.
- Because AWS doesn't maintain third-party repositories, we can't guarantee that they contain the latest version of the AWS CLI.

macOS version support matrix

AWS CLI version	Supported macOS version
2.21.0 – current	11+
2.17.0 – 2.20.0	10.15+
2.0.0 – 2.16.12	10.14 and below

Install or update the AWS CLI

If you are updating to the latest version, use the same installation method that you used in your current version. You can install the AWS CLI on macOS in the following ways.

Command line installer - All users

If you have `sudo` permissions, you can install the AWS CLI for all users on the computer. We provide the steps in one easy to copy and paste group. See the descriptions of each line in the following steps.

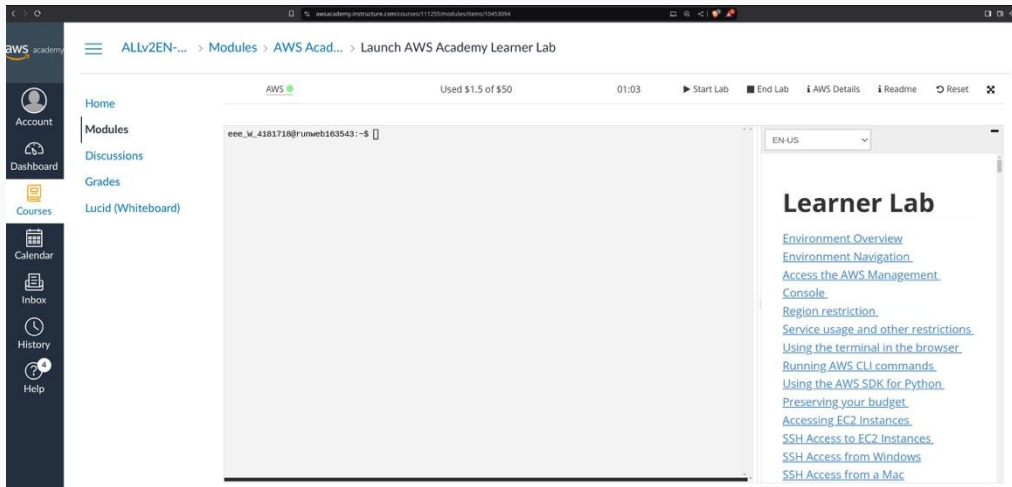
```
$ curl "https://awscli.amazonaws.com/AWSCLIV2.pkg" -o "AWSCLIV2.pkg"
$ sudo installer -pkg AWSCLIV2.pkg -target /
```

Guided installation instructions

- Download the file using the `curl` command. The `-o` option specifies the file name that the downloaded package is written to. In this example, the file is written to `AWSCLIV2.pkg` in the current folder.

Configuring the AWS Credentials

We're now going to visit the Learner Lab page on the AWS Academy website to get our credentials. You have [this guide](#) and [this guide](#) available on the subject's [website](#) to help you with setting up AWS. Wait until the lab loads and you see the page below.



The screenshot shows the AWS Academy Learner Lab interface. The browser address bar displays `awsacademy.instructure.com/courses/111255/modules/items/70403094`. The page title is "Launch AWS Academy Learner Lab". The interface includes a sidebar with navigation links: Home, Modules, Discussions, Grades, Lucid (Whiteboard), Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area shows a terminal window with the prompt `eee_w_4181718@runweb163543:~$`. The top status bar indicates "AWS" with a green dot, "Used \$1.5 of \$50", and a timer "01:03". Action buttons include "Start Lab", "End Lab", "AWS Details", "Readme", "Reset", and a close button. On the right, the "Learner Lab" section lists several links: [Environment Overview](#), [Environment Navigation](#), [Access the AWS Management Console](#), [Region restriction](#), [Service usage and other restrictions](#), [Using the terminal in the browser](#), [Running AWS CLI commands](#), [Using the AWS SDK for Python](#), [Preserving your budget](#), [Accessing EC2 Instances](#), [SSH Access to EC2 Instances](#), [SSH Access from Windows](#), and [SSH Access from a Mac](#).

Configuring the AWS Credentials

Now click on **AWS Details** and then on **Show** to reveal your credentials.

The screenshot shows the AWS Academy Learner Lab interface. The breadcrumb navigation at the top reads: **ALLv2EN-...** > **Modules** > **AWS Acad...** > **Launch AWS Academy Learner Lab**. The main header bar displays the **AWS** logo, **Used \$1.6 of \$50**, a timer at **01:01**, and buttons for **Start Lab**, **End Lab**, **AWS Details** (highlighted with a red circle 1), **Readme**, **Reset**, and a close icon. On the left sidebar, the **Modules** section is expanded, showing **Home**, **Modules**, **Discussions**, **Grades**, and **Lucid (Whiteboard)**. The central terminal area shows the prompt `eee_w_4181718@runweb163552:~$`. On the right, the **Cloud Access** panel is open, showing **AWS CLI:** with a **Show** button (highlighted with a red circle 2). Below this, the **Cloud Labs** section displays session information: **Remaining session time: 01:00:53(61 minutes)**, **Session started at: 2025-03-03T08:48:13-0800**, **Session to end at: 2025-03-03T12:48:13-0800**, and **Accumulated lab time: 1 day 05:41:00 (1781 minutes)**.

Configuring the AWS Credentials

You'll see some text containing your credentials.

Used \$2.1 of \$5003:46▶ Start Lab■ End Labi AWS Detailsi Readme↻ Reset✕

Cloud AccessClose

AWS CLI:
Copy and paste the following into ~/.aws/credentials

[default]
aws_access_key_id=ASIA2CKYVHJA03I2XRZ3
aws_secret_access_key=dudm/D3hbusUkb6iqzEXFcVXjjQkghDy3+ALE0a3
aws_session_token=IQoJb3JpZ2luX2VjELf////////wEaCXVzLXdlc3QtMiJGMEQCIE00uLiVzFVLHv5EbZPZj8hPqPJGou84lGfM0ezfiSP0AiB+S2vrdPaz4TFMEYXwe1e0Hr/C05m0B3N72MZHQ1n0DyqzAgjw////////8BEAEaDDY5MjIxMjU0NjExMiIMCDFeGSKhijJKhgHCKocCd6djgX83VD2PG83XsoMK972gcLP6P8T0ZvujiY4RT0S2uqbjAW/2cNwdpukyH8LX0zAjI00CzpwR+M5HrC/7vUPZSANonVQdJvmbBrXZ+ln8SzzTuBFq1zD6bRhnV7iHhA8naNoXe4eNAM0C09HQ6JmfVAoPTQ3nGDPGKEfIPmG6KzIR3H0vmuZSvsRPb4MctQxk/x4m1d4KuJ6lDeEWobNnd7fq1kvFfi+C/Gvc5

Configuring the AWS Credentials

Next we need to paste that text to a file called `credentials` inside the `.aws` folder in our home directory.

Make sure the `.aws` folder exists on your local machine by running `mkdir .aws` command (remember if it throws an error there's nothing to worry about, it just means the folder already exists). Now we are going to create the `credentials` file inside the `.aws` folder. Run the following command:

```
notepad .aws/credentials.
```

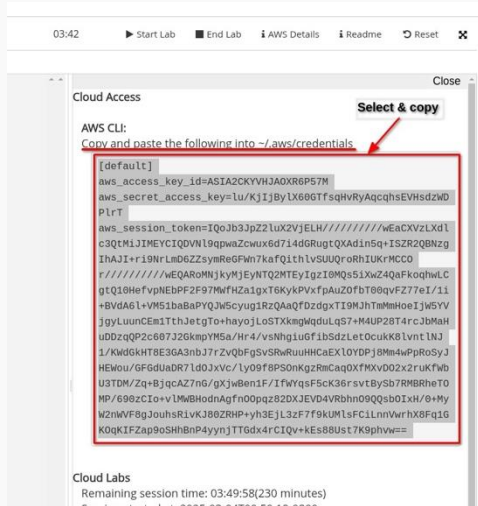
or for MacOS users:

```
open .aws/credentials.
```

This will open a text editor where you can write the credentials.

Configuring the AWS Credentials

Go back to the AWS Academy website and copy the text containing your credentials.



The screenshot shows a web interface for 'Cloud Access' with a 'Close' button in the top right. Below the title, there are navigation links: 'Start Lab', 'End Lab', 'AWS Details', 'Readme', 'Reset', and a close icon. The main content area is titled 'AWS CLI:' and contains the instruction 'Copy and paste the following into ~/.aws/credentials'. A red box highlights the following text, and a red arrow points to it from the 'Select & copy' button. At the bottom, there is a 'Cloud Labs' section showing 'Remaining session time: 03:49:58(230 minutes)'.

03:42 ▶ Start Lab ■ End Lab ⓘ AWS Details ⓘ Readme ↺ Reset ✕

Cloud Access Close

Select & copy

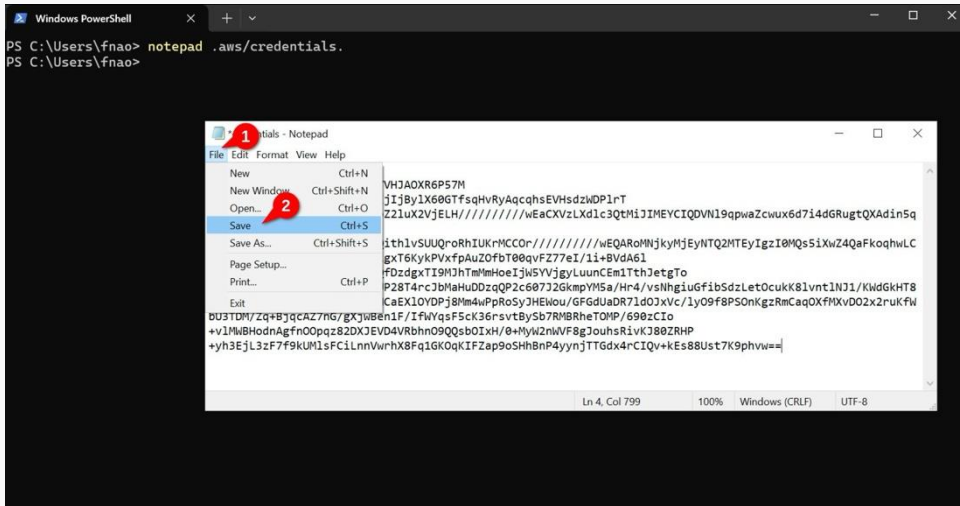
AWS CLI:
Copy and paste the following into ~/.aws/credentials

```
[default]
aws_access_key_id=ASIA2CKYVHJAOXR6P57M
aws_secret_access_key=lu/KjIjByLx60GTfsqHvRyAqcqhsEVHsdzWD
PlrT
aws_session_token=IQoJb3JpZ2luX2VjELH////////wEaCXVzLXdl
c3QtMiJIMEYCIQDVNl9qpwaZcwux6d7i4dGRugtQXAdin5q+ISZR2QBnzg
IhAJI+ri9NrLmd6ZZsymReGFwn7kaFqithlvSUUQroRhIUkrMCCO
r////////wEQARoMNjkyMjEyNTQ2MTEyIgzI0MQs5iXwZ4QaFkoqhWLC
gtQ10HefvpNEbPF2F97MwfHza1gxT6KykPVxfpAuZ0fbT00qvFZ77eI/iI
+BVdA6l+VM51baBaPYQJw5cyug1RzQAaQfDzdgxTI9MJhTmMmHoeIjW5YV
jgyLuunCEm1TthJetgTo+hayojLoSTXkmgWqduLqS7+M4UP28T4rcJbMaH
uDDzqQP2c607J2GkmpYM5a/Hr4/vsNhgIuGf1bSdzLet0cukK8lvntlNJ
1/KWdGkHT8E3GA3nbJ7rZvQbFgSvSRwRuuHHCaEXLOYDPj8Mm4wPpRoSyJ
HEWou/6FGdUaDR7ld0Jxvc/ly09f8PS0nKgZrmCaQ0XfMXvD02x2ruKfWb
U3TDM/Zq+BjqcAZ7n6/gXjwBen1F/IfwYqsF5cK36rsvtBySb7RMBRheTO
MP/690zCIo+vLMwBHodnAgfn00pqz82DXJEVD4VRbhn09QQsb0IxH/0+My
W2nWVF8gJouhsRlvKJ80ZRHP+yh3EjL3zF7f9kUmlsFC1LnnVvrhX8Fq1G
K0qKIFZap9oSHhBnP4yynjTTGdx4rCIQv+kEs88Ust7K9phvw==
```

Cloud Labs
Remaining session time: 03:49:58(230 minutes)
Session started at 2025-02-04T00:00:00.000Z

Configuring the AWS Credentials

Now go back to the text editor. Paste the credentials and save the file. You can now exit the text editor.



Configuring the AWS Credentials

We can check the contents of the file using `cat`:

```
PS C:\Users\fnao> cat .aws\credentials
[default]
aws_access_key_id=ASIA2CKYVHJAOXR6P57M
aws_secret_access_key=lu/KjIjBylX60GTfsqHvRyAqcqhsEVHsdzWDPlrT
aws_session_token=IQoJb3JpZ2...
PS C:\Users\fnao>
```

Configuring the AWS Credentials

To test if the configuration was successful, run `aws sts get-caller-identity` and you should see something like this:

```
PS C:\Users\fnao> aws sts get-caller-identity
{
  "UserId": "AROA2CKYVHJALK46ZMHVM:user3869188=Ferran_Aran_Test",
  "Account": "692212546112",
  "Am": "am:aws:sts::692212546112:assumed-role/voclabs/user3869188=Ferran_Aran_Test"
}

PS C:\Users\fnao>
```

Great! We can now run AWS CLI commands on our local machine to manage our AWS resources. For example, we can upload files to an S3 bucket.

S3 Buckets

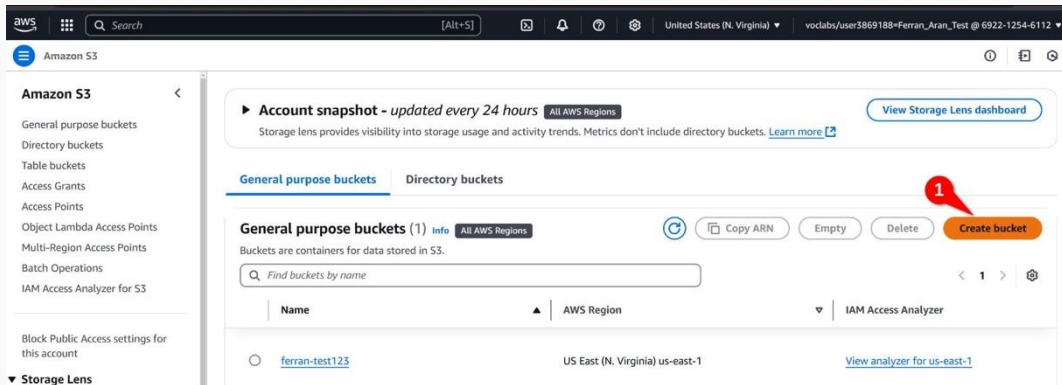
Creating a S3 bucket

Use the searchbar to head to the S3 service.



Creating a S3 bucket

Click on `Create bucket`.



The screenshot shows the Amazon S3 console interface. On the left is a navigation sidebar with the 'Amazon S3' header and a list of options: General purpose buckets, Directory buckets, Table buckets, Access Grants, Access Points, Object Lambda Access Points, Multi-Region Access Points, Batch Operations, IAM Access Analyzer for S3, Block Public Access settings for this account, and Storage Lens. The main content area has a top banner for 'Account snapshot - updated every 24 hours' and a 'View Storage Lens dashboard' button. Below this are tabs for 'General purpose buckets' (selected) and 'Directory buckets'. The 'General purpose buckets' section shows 'General purpose buckets (1)' with an 'Info' link and 'All AWS Regions' filter. It includes a search bar 'Find buckets by name', a refresh icon, and buttons for 'Copy ARN', 'Empty', 'Delete', and 'Create bucket'. A red circle with the number '1' points to the 'Create bucket' button. Below the buttons is a table with columns 'Name', 'AWS Region', and 'IAM Access Analyzer'. The table contains one entry: a radio button, the bucket name 'ferran-test123', the region 'US East (N. Virginia) us-east-1', and a link 'View analyzer for us-east-1'.

aws [Search] [Alt+S] United States (N. Virginia) voclabs/user3869188=Ferran_Aran_Test @ 6922-1254-6112

Amazon S3

Amazon S3

- General purpose buckets
- Directory buckets
- Table buckets
- Access Grants
- Access Points
- Object Lambda Access Points
- Multi-Region Access Points
- Batch Operations
- IAM Access Analyzer for S3

Block Public Access settings for this account

▼ Storage Lens

► **Account snapshot - updated every 24 hours** All AWS Regions View Storage Lens dashboard

Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets Directory buckets

General purpose buckets (1) Info All AWS Regions

Buckets are containers for data stored in S3.

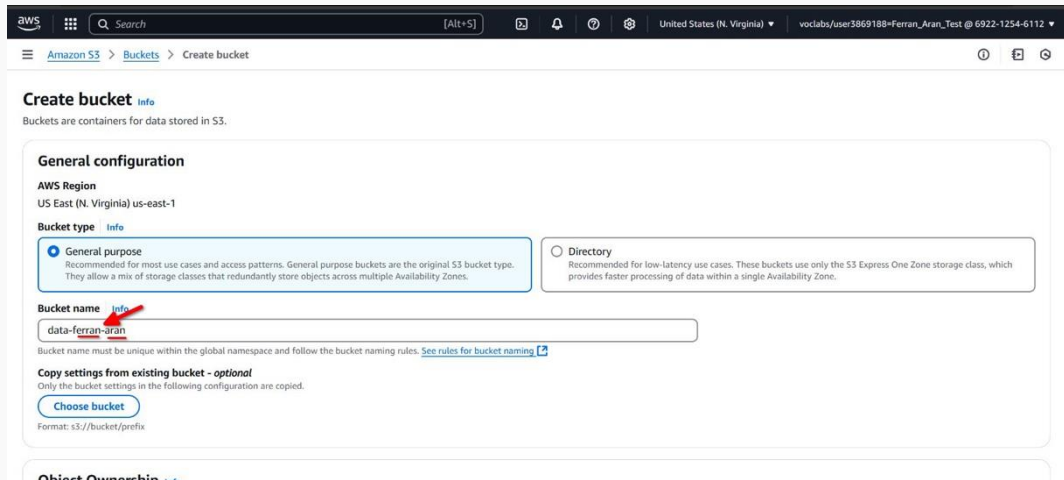
Find buckets by name

< 1 > ⚙

Name	AWS Region	IAM Access Analyzer
☐ ferran-test123	US East (N. Virginia) us-east-1	View analyzer for us-east-1

Creating a S3 bucket

Name the bucket `data-{your-name}`. For example I will be naming it `data-ferran-aran`. **Bucket names have to be unique across all of AWS.**



The screenshot shows the AWS Management Console interface for creating a new S3 bucket. The top navigation bar includes the AWS logo, a search bar, and various utility icons. The breadcrumb trail indicates the path: Amazon S3 > Buckets > Create bucket. The main heading is 'Create bucket' with an 'info' link. Below this, a sub-header 'General configuration' is present. The 'AWS Region' is set to 'US East (N. Virginia) us-east-1'. The 'Bucket type' section has two options: 'General purpose' (selected with a radio button) and 'Directory'. The 'General purpose' option is described as recommended for most use cases and allows a mix of storage classes. The 'Directory' option is recommended for low-latency use cases. The 'Bucket name' field contains the text 'data-ferran-aran', with a red arrow pointing to it. Below the field, a note states that the bucket name must be unique and follows specific naming rules, with a link to 'See rules for bucket naming'. There is also a section for 'Copy settings from existing bucket - optional' with a 'Choose bucket' button. At the bottom, the 'Object Ownership' section is partially visible.

Create bucket [info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [info](#)

data-ferran-aran

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [info](#)

Creating a S3 bucket

Leave everything else as default, scroll all the way down and click on **Create bucket**.

Amazon S3 > Buckets > Create bucket

ⓘ ⓘ ⓘ

Add tag

Default encryption ⓘ

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type ⓘ

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#). ⓘ

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#) ⓘ

☐ Disable
☒ Enable

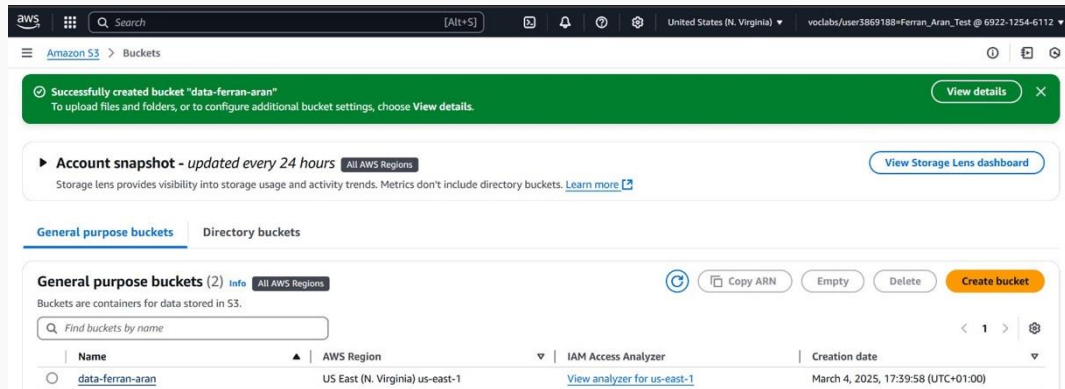
▶ **Advanced settings**

ⓘ After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel **Create bucket**

Creating a S3 bucket

You will now be headed to the S3 dashboard. You should see a success message and the bucket you just created.



The screenshot shows the AWS S3 console interface. At the top, the AWS logo and navigation bar are visible, including a search bar and region selector set to 'United States (N. Virginia)'. Below the navigation bar, the breadcrumb 'Amazon S3 > Buckets' is shown. A prominent green success banner at the top of the main content area states: 'Successfully created bucket "data-ferran-aran"'. Below this, a section for 'Account snapshot' is visible. The main content area is divided into two tabs: 'General purpose buckets' (selected) and 'Directory buckets'. Under the 'General purpose buckets' tab, there is a sub-header 'General purpose buckets (2)' with an 'Info' icon and a filter 'All AWS Regions'. A description states 'Buckets are containers for data stored in S3.' Below this is a search bar 'Find buckets by name'. To the right of the search bar are buttons for 'Copy ARN', 'Empty', 'Delete', and a prominent orange 'Create bucket' button. Below these buttons is a table listing the buckets. The table has columns for 'Name', 'AWS Region', 'IAM Access Analyzer', and 'Creation date'. One bucket is listed: 'data-ferran-aran' in the 'US East (N. Virginia) us-east-1' region. The 'IAM Access Analyzer' column for this bucket contains a link 'View analyzer for us-east-1'. The 'Creation date' is 'March 4, 2025, 17:39:58 (UTC+01:00)'. The table is paginated to show 1 item.

aws [Search] [Alt+S] [Icons] United States (N. Virginia) ▼ voclabs/user3869188=Ferran_Aran_Test @ 6922-1254-6112 ▼

Amazon S3 > Buckets

Successfully created bucket "data-ferran-aran"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

Account snapshot - updated every 24 hours [All AWS Regions](#) [View Storage Lens dashboard](#)
Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets | Directory buckets

General purpose buckets (2) [Info](#) [All AWS Regions](#)
Buckets are containers for data stored in S3.

Find buckets by name

Copy ARN Empty Delete Create bucket

Name	AWS Region	IAM Access Analyzer	Creation date
data-ferran-aran	US East (N. Virginia) us-east-1	View analyzer for us-east-1	March 4, 2025, 17:39:58 (UTC+01:00)

Uploading a file to the bucket

From the S3 dashboard

1. Click on the bucket.
2. Create a folder and name it *test*.
3. Click on the folder.
4. Click on Upload.
5. Click on Add files and select an example file.
6. Click Upload.

Our bucket **data-{your-name}**/ now contains one file. It is stored in **s3://data-{your-name}/test**. This file is private by default. Only the owner can access them.

From the AWS CLI

- `aws s3 cp <local-file-path> s3://<bucket-name>/<optional-target-name>`

Example (upload photo.jpg to a bucket named my-bucket):

```
aws s3 cp photo.jpg s3://my-bucket/
```

Sync with a bucket

Syncing a local directory to upload to a bucket

From the AWS CLI

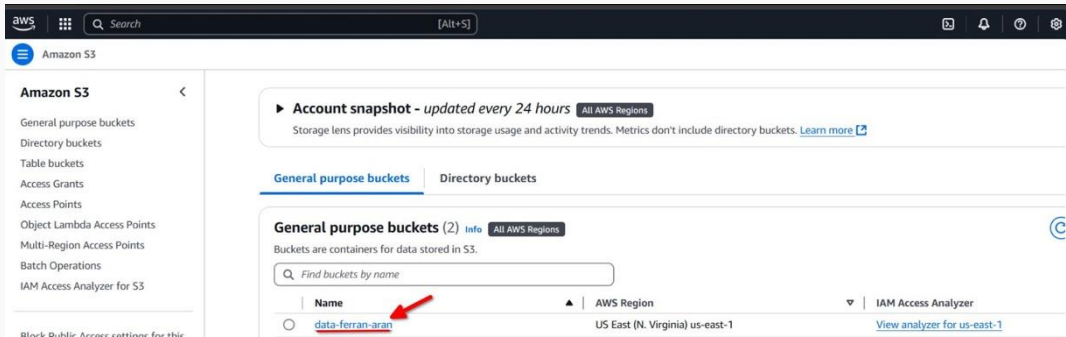
1. Go to your desktop and create a folder named `data`.
2. Create a file named `my-dataset.txt` and write some text in it. Save it.
3. Open a terminal and navigate to the Desktop directory. Use `cd Desktop`.
4. Run `aws s3 sync data s3://data-{your-name}` to upload the files to the bucket. In my case this is the result:

```
PS C:\Users\fnao\Desktop> aws s3 sync data s3://data-ferran-aran
upload: data\my-dataset.txt to s3://data-ferran-aran/my-dataset.txt
```

That's it! You have now uploaded a file to the bucket!

Inspecting the bucket from the AWS Console

Go to the S3 dashboard on AWS and click on the bucket we just created.

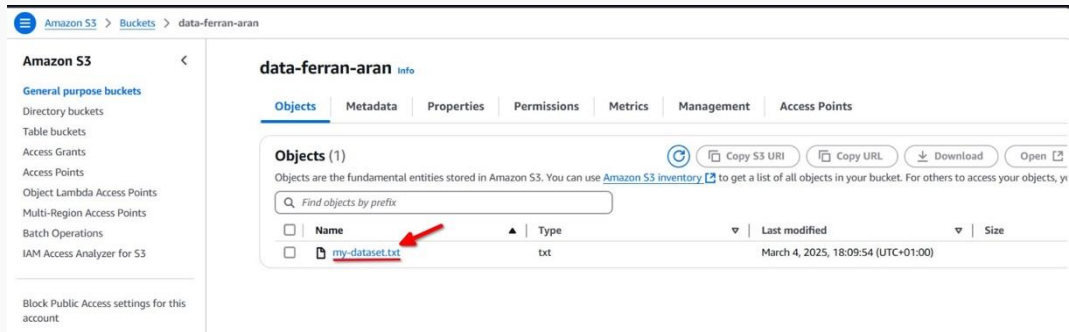


The screenshot shows the AWS S3 console interface. On the left is a navigation sidebar with the 'Amazon S3' header and a list of options: General purpose buckets, Directory buckets, Table buckets, Access Grants, Access Points, Object Lambda Access Points, Multi-Region Access Points, Batch Operations, and IAM Access Analyzer for S3. The main content area has a top section for 'Account snapshot - updated every 24 hours' with a 'Learn more' link. Below this are two tabs: 'General purpose buckets' (selected) and 'Directory buckets'. The 'General purpose buckets' section shows 'Buckets are containers for data stored in S3.' and a search bar 'Find buckets by name'. A table lists the buckets with columns for Name, AWS Region, and IAM Access Analyzer. The first bucket, 'data-ferran-aran', is highlighted with a red arrow. The table also shows the region 'US East (N. Virginia) us-east-1' and a link to 'View analyzer for us-east-1'.

Name	AWS Region	IAM Access Analyzer
data-ferran-aran	US East (N. Virginia) us-east-1	View analyzer for us-east-1

Inspecting the bucket from the AWS Console

You will see the contents of the bucket, in this case the file `my-dataset.txt`.



The screenshot shows the AWS S3 console interface. On the left, the 'Amazon S3' sidebar is visible with a list of navigation options. The main area displays the 'data-ferran-aran' bucket. The 'Objects' tab is active, showing a table of objects. A red arrow points to the filename 'my-dataset.txt' in the table.

Amazon S3 <

- General purpose buckets
- Directory buckets
- Table buckets
- Access Grants
- Access Points
- Object Lambda Access Points
- Multi-Region Access Points
- Batch Operations
- IAM Access Analyzer for S3

data-ferran-aran Info

Objects Metadata Properties Permissions Metrics Management Access Points

Objects (1)   Copy S3 URI  Copy URL  Download  Open

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you can use [Access Points](#).

☐	Name	Type	Last modified	Size
☐	 <u>my-dataset.txt</u>	txt	March 4, 2025, 18:09:54 (UTC+01:00)	

Block Public Access settings for this account

Inspecting the bucket from the AWS Console

Click on the file to see further details.

The screenshot displays the AWS Management Console interface for an Amazon S3 bucket named 'data-ferran-aran'. The breadcrumb navigation shows the path: Amazon S3 > Buckets > data-ferran-aran > my-dataset.txt. The left-hand navigation pane lists various S3 features, including 'General purpose buckets' and 'Storage Lens'. The main content area is titled 'my-dataset.txt' and includes tabs for 'Properties', 'Permissions', and 'Versions'. The 'Properties' tab is active, showing an 'Object overview' section with details such as the owner (aws-labs), region (US East (N. Virginia)), last modified date (March 4, 2025), size (18.0 B), type (txt), and key (my-dataset.txt). On the right side, there are buttons for 'Copy S3 URI', 'Download', 'Open', and 'Object actions'. Below these buttons, the 'S3 URI' is displayed as <s3://data-ferran-aran/my-dataset.txt>, which is underlined and highlighted by a red arrow. Other details shown include the Amazon Resource Name (ARN), Entity tag (Etag), and Object URL.

Amazon S3

General purpose buckets

Directory buckets

Table buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

▼ Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

my-dataset.txt

Properties Permissions Versions

Object overview

Owner
aws-labs0w4907751t1669516777

AWS Region
US East (N. Virginia) us-east-1

Last modified
March 4, 2025, 18:09:54 (UTC+01:00)

Size
18.0 B

Type
txt

Key
my-dataset.txt

Copy S3 URI Download Open Object actions

S3 URI
<s3://data-ferran-aran/my-dataset.txt>

Amazon Resource Name (ARN)
<arn:aws:s3:::data-ferran-aran/my-dataset.txt>

Entity tag (Etag)
[0fee9acadfe8753f721d44b985a90c67](#)

Object URL
<https://data-ferran-aran.s3.us-east-1.amazonaws.com/my-dataset.txt>

Working with S3 from python

Configuring AWS credentials on an EC2 instance

The first thing we'll have to do is to connect to our EC2 instance we configured during last session. More information on last session can be found [here \(Session 3\)](#).

Once we are connected through SSH on a remote terminal, we'll need to configure AWS Credentials similarly to how we did on our local machine. But this time we will have to use a terminal editor.

If this steps get confusing I suggest checking [this guide](#) on the subject's website for a more detailed explanation.

Configuring AWS credentials on an EC2 instance

EC2 machines come with AWS CLI already installed so we won't have to worry about that.

Remember we first need to make sure the `.aws` folder exists in the home directory of the user we are using. If it doesn't exist we can create it by running `mkdir .aws`.

Next we need to create the `credentials` file inside the `.aws` folder. We can do this by running `nano .aws/credentials`.

The nano text editor will open and we can paste the credentials we copied from the AWS Academy website.

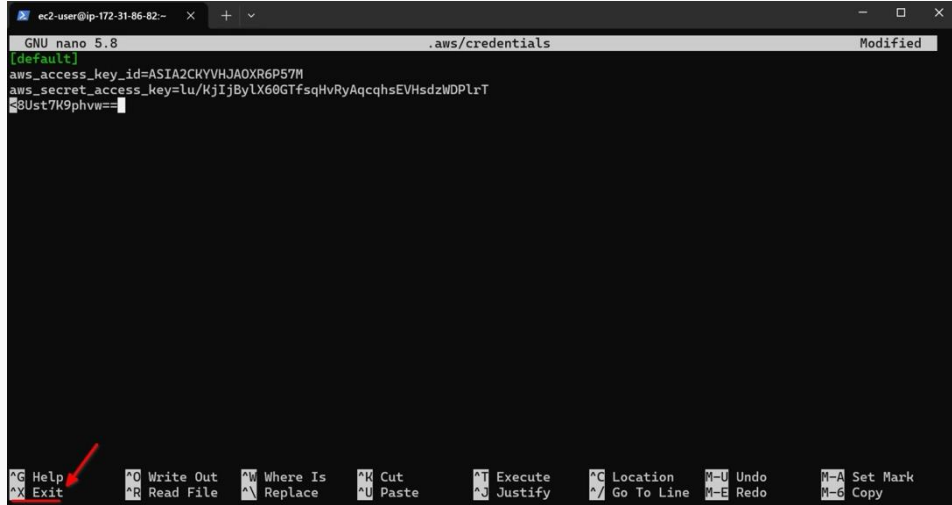
To save the file we can press `Ctrl + X` and then `Y` and finally `Enter`. See the following screenshots of the process.

Configuring AWS credentials on an EC2 instance

The screenshot shows a terminal window with the GNU nano 5.8 text editor. The editor is open to a file named `.aws/credentials`. The terminal title bar indicates the user is `ec2-user` on a machine with IP `172-31-86-82`. The bottom status bar displays the following keyboard shortcuts:

<code>^G</code> Help	<code>^O</code> Write Out	<code>^W</code> Where Is	<code>^K</code> Cut	<code>^T</code> Execute	<code>^C</code> Location	<code>^M-U</code> Undo	<code>^M-A</code> Set Mark
<code>^X</code> Exit	<code>^R</code> Read File	<code>^_\</code> Replace	<code>^U</code> Paste	<code>^J</code> Justify	<code>^/</code> Go To Line	<code>^M-E</code> Redo	<code>^M-G</code> Copy

Configuring AWS credentials on an EC2 instance



The screenshot shows a terminal window titled "ec2-user@ip-172-31-86-82:~". The terminal is running the GNU nano 5.8 editor, editing the file ".aws/credentials". The file content is as follows:

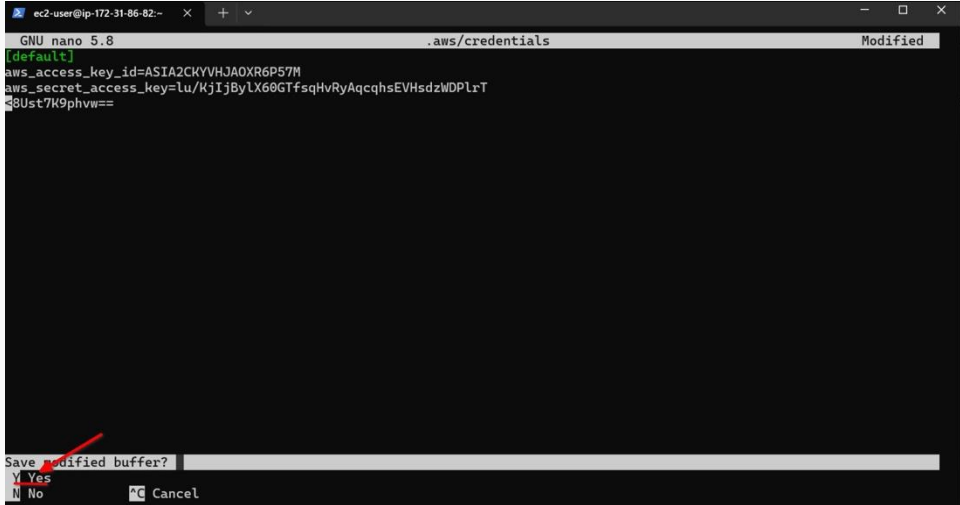
```
[default]
aws_access_key_id=ASIA2CKYVHJA0XR6P57M
aws_secret_access_key=lu/KjIjBylX60GTfsqHvRyAqcqhsEVHsdzWDPlrT
8Ust7K9phvw==
```

The bottom of the terminal displays the nano editor's help menu, which includes the following options:

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^_\ Replace	^U Paste	^J Justify	^/ Go To Line	M-E Redo	M-6 Copy

A red arrow points to the **^X Exit** option in the help menu.

Configuring AWS credentials on an EC2 instance

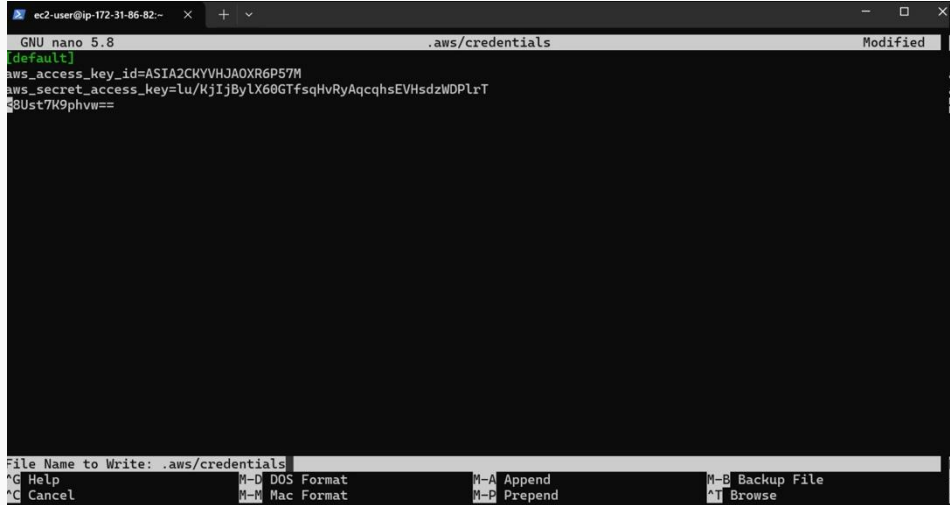


The screenshot shows a terminal window with a nano editor editing a file named `.aws/credentials`. The editor's status bar at the top indicates "GNU nano 5.8" and "Modified". The file content contains the following text:

```
[default]
aws_access_key_id=ASIA2CKYVHJAOXR6P57M
aws_secret_access_key=lu/KjIjBylX60GTfsqHvRyAqcqhsEVHsdzWDP1rT
8Ust7K9phvw==
```

At the bottom of the terminal, a prompt asks "Save modified buffer?". A red arrow points to the "Y" option, which is selected. The other options are "N" for "No" and "^C" for "Cancel".

Configuring AWS credentials on an EC2 instance



The screenshot shows a terminal window with a nano editor editing a file named `.aws/credentials`. The terminal title bar indicates the user is `ec2-user` on an instance with IP `ip-172-31-86-82`. The nano editor's status bar at the top shows `GNU nano 5.8`, the file path `.aws/credentials`, and the state `Modified`. The editor content shows the following text:

```
[default]
aws_access_key_id=ASIA2CKYVHJA0XR6P57M
aws_secret_access_key=lu/KjIjBylX60GTfsqHvRyAqcqhsEVHsdzWDPlrT
s8Ust7K9phvw==
```

The bottom of the terminal shows the nano editor's command menu:

```
File Name to Write: .aws/credentials
^G Help      M-D DOS Format  M-A Append     M-B Backup File
^C Cancel    M-M Mac Format  M-P Prepend    ^T Browse
```

Configuring AWS credentials on an EC2 instance

We can now test if the configuration was successful by running `aws sts get-caller-identity`.

If the configuration was successful we should see something like this:

```
{  
  "UserId": "AROA2CKYVHJALK46ZMHVM:user3869188=Ferran_Aran_Test",  
  "Account": "692212546112",  
  "Arn": "arn:aws:sts::692212546112:assumed-role/voclabs/user3869188=Ferran_Aran_Test"  
}
```

Reading files from S3 with python

We are going to use the *boto3 python library* to access and write S3 files from a jupyter notebook. This library allows Python developers to write software that makes use of services like Amazon S3 on AWS.

We'll need to first activate one of the environment we created during the last session. For example, I will be using `project1` environment. Follow the steps below to activate the environment and install `boto3`:

```
cd project1
source .project1/bin/activate
pip install boto3
```


Once that is done, launch the jupyter server with the command below:

```
jupyter notebook --no-browser --port=8888 --ip=0.0.0.0
```

Remember on last session we saw the steps to access the jupyter notebook from our local machine. If you need a refresher you can check [Session 3](#) on the subject's website.

Reading files from S3 with python

Open a jupyter notebook and paste the following code:

```
import boto3

DATA_BUCKET_NAME = "data-your-name"
DATA_FILE_NAME = "my-dataset.txt" # Path to the file in S3

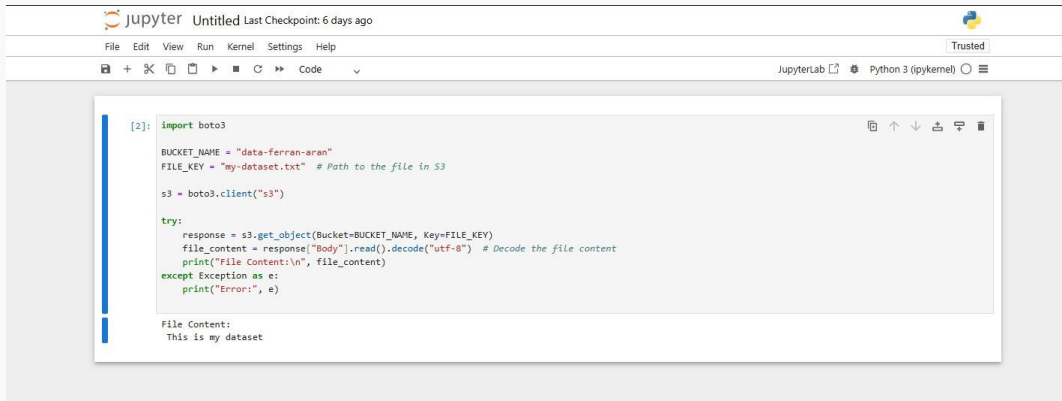
s3 = boto3.client("s3")

response = s3.get_object(Bucket=DATA_BUCKET_NAME, Key=DATA_FILE_NAME)
file_content = response["Body"].read().decode("utf-8") # Decode the file content
print("File Content:\n", file_content)
```

Be careful to replace `data-your-name` with the name of the bucket you created and `my-dataset.txt` with the name of the file you uploaded.

Reading files from S3 with python

If everything is correct you should see the content of the file printed in the notebook.



The image shows a JupyterLab interface. The top bar includes the Jupyter logo, the text "jupyter", and "Untitled Last Checkpoint: 6 days ago". On the right is a Python logo and a "Trusted" button. Below this is a menu bar with "File", "Edit", "View", "Run", "Kernel", "Settings", and "Help". A toolbar contains icons for file operations and execution. The right side of the top bar shows "JupyterLab", a gear icon, "Python 3 (ipykernel)", and a hamburger menu icon. The main area contains a code cell with the following Python code:

```
[2]: import boto3

BUCKET_NAME = "data-ferran-aran"
FILE_KEY = "my-dataset.txt" # Path to the file in S3

s3 = boto3.client("s3")

try:
    response = s3.get_object(Bucket=BUCKET_NAME, Key=FILE_KEY)
    file_content = response["Body"].read().decode("utf-8") # Decode the file content
    print("File Content:\n", file_content)
except Exception as e:
    print("Error:", e)
```

Below the code cell, the output is displayed:

```
File Content:
This is my dataset
```

RECAP: Accessing the file from the notebook

To access the file from the notebook we have used our aws-credentials for the current user (owner of the bucket).

Be aware!

Someone could log into your jupyter instance in the browser and access the file using your credentials.

What to do?

In real life, we would use IAM roles to give the notebook the necessary permissions to access the file.

But, in AWS Educate, we can not use IAM roles.

Another option

Make the file public and access it without credentials :)

Writing files to S3 with python

We can also write files to S3 using boto3. Lets first create another bucket to store the files we will write from the notebook. This one we will call `results-{your-name}`.

Leave everything as default like before and scroll all the way down to click on `Create bucket`.

The screenshot shows the AWS Management Console interface for creating a new S3 bucket. The top navigation bar includes the AWS logo, a search bar, and user information. The breadcrumb trail indicates the path: Amazon S3 > Buckets > Create bucket. The main heading is 'Create bucket' with an 'Info' link. Below this, a note states 'Buckets are containers for data stored in S3.' The 'General configuration' section is expanded, showing the 'AWS Region' as 'US East (N. Virginia) us-east-1'. Under 'Bucket type', the 'General purpose' option is selected with a radio button, while the 'Directory' option is unselected. The 'General purpose' description recommends it for most use cases and access patterns. The 'Bucket name' field contains the text 'results-ferran-aran' and is underlined in red. A note below the field states that the bucket name must be unique and follow naming rules, with a link to 'See rules for bucket naming'. At the bottom, there is a section for 'Copy settings from existing bucket - optional' with a 'Choose bucket' button and a format example 'Format: s3://bucket/prefix'.

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)
results-ferran-aran

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Writing files to S3 with python

We're now going to use the following code to write a file to the bucket we just created:

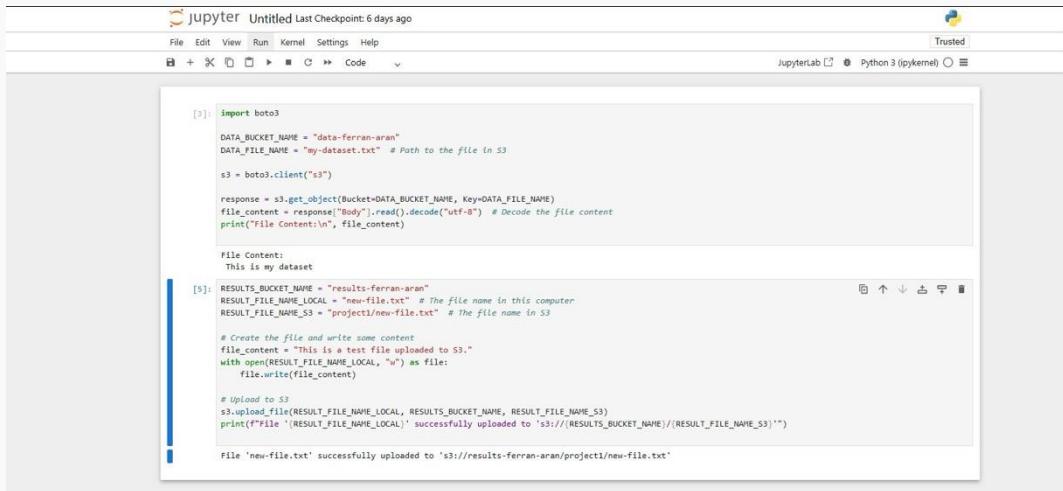
```
RESULTS_BUCKET_NAME = "results-your-name"
RESULT_FILE_NAME_LOCAL = "new-file.txt" # The file name in this computer
RESULT_FILE_NAME_S3 = "project1/new-file.txt" # The file name in S3

# Create the file and write some content
file_content = "This is a test file uploaded to S3."
with open(RESULT_FILE_NAME_LOCAL, "w") as file:
    file.write(file_content)

# Upload to S3
s3.upload_file(RESULT_FILE_NAME_LOCAL, RESULTS_BUCKET_NAME, RESULT_FILE_NAME_S3)
print(f"File '{RESULT_FILE_NAME_LOCAL}' successfully uploaded to 's3://{RESULTS_BUCKET_NAME}/{RESULT_FILE_NAME_S3}'")
```

Writing files to S3 with python

If everything is correct you should see the following:



```
[3]: import boto3

DATA_BUCKET_NAME = "data-ferran-aran"
DATA_FILE_NAME = "my-dataset.txt" # Path to the file in S3

s3 = boto3.client("s3")

response = s3.get_object(Bucket=DATA_BUCKET_NAME, Key=DATA_FILE_NAME)
file_content = response["Body"].read().decode("utf-8") # Decode the file content
print("File Content:\n", file_content)

File Content:
This is my dataset

[5]: RESULTS_BUCKET_NAME = "results-ferran-aran"
RESULT_FILE_NAME_LOCAL = "new-file.txt" # The file name in this computer
RESULT_FILE_NAME_S3 = "project1/new-file.txt" # The file name in S3

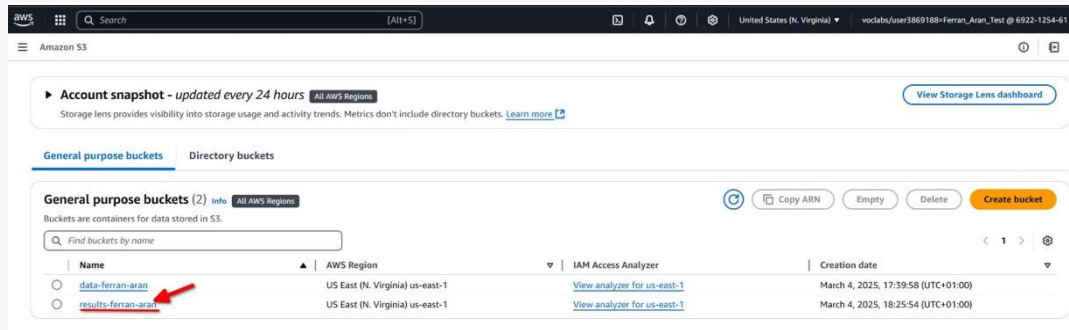
# Create the file and write some content
file_content = "This is a test file uploaded to S3."
with open(RESULT_FILE_NAME_LOCAL, "w") as file:
    file.write(file_content)

# Upload to S3
s3.upload_file(RESULT_FILE_NAME_LOCAL, RESULTS_BUCKET_NAME, RESULT_FILE_NAME_S3)
print(f"File '{RESULT_FILE_NAME_LOCAL}' successfully uploaded to 's3://{RESULTS_BUCKET_NAME}/{RESULT_FILE_NAME_S3}'")

File 'new-file.txt' successfully uploaded to 's3://results-ferran-aran/project1/new-file.txt'
```

Inspecting the bucket from the AWS Console

We can now navigate to the S3 dashboard and click on the `results-{your-name}` bucket.

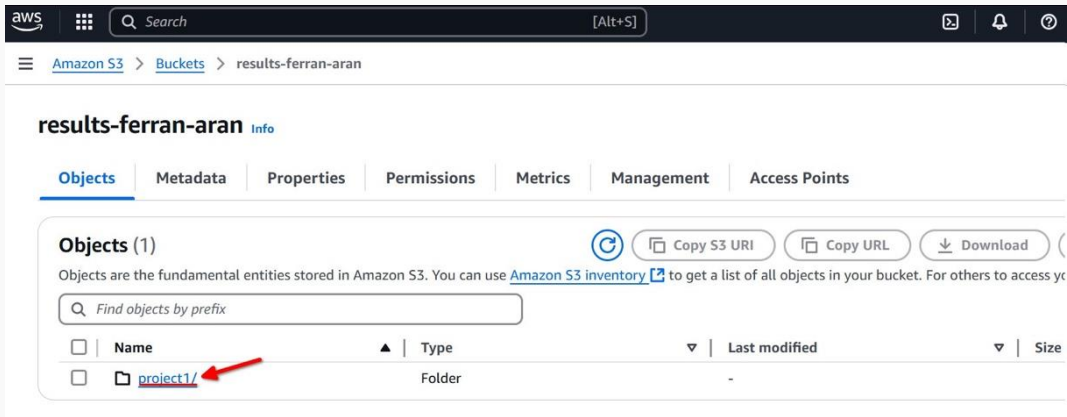


The screenshot shows the AWS S3 console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and the region 'United States (N. Virginia)'. Below this, the 'Amazon S3' header is visible. The main content area shows an 'Account snapshot' and a 'View Storage Lens dashboard' button. Under the 'General purpose buckets' tab, there's a section titled 'General purpose buckets (2)' with a search bar and a table of buckets. A red arrow points to the 'results-ferran-aran' bucket in the table.

Name	AWS Region	IAM Access Analyzer	Creation date
data-ferran-aran	US East (N. Virginia) us-east-1	View analyzer for us-east-1	March 4, 2025, 17:39:58 (UTC+01:00)
results-ferran-aran	US East (N. Virginia) us-east-1	View analyzer for us-east-1	March 4, 2025, 18:25:54 (UTC+01:00)

Inspecting the bucket from the AWS Console

Inside the bucket we should see the folder we just created, click on it.



The screenshot shows the AWS S3 console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and utility icons. Below the navigation bar, the breadcrumb trail reads 'Amazon S3 > Buckets > results-ferran-aran'. The main heading is 'results-ferran-aran' with an 'Info' link. A tabbed interface shows 'Objects' as the active tab, with other tabs for 'Metadata', 'Properties', 'Permissions', 'Metrics', 'Management', and 'Access Points'. Under the 'Objects' tab, there's a section titled 'Objects (1)' with a refresh icon and three buttons: 'Copy S3 URI', 'Copy URL', and 'Download'. A descriptive text states: 'Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access yo'. Below this is a search bar with the placeholder 'Find objects by prefix'. A table lists the objects with columns: 'Name', 'Type', 'Last modified', and 'Size'. The first row shows a folder icon, the name 'project1/' (underlined and highlighted with a red arrow), the type 'Folder', and a hyphen in the 'Last modified' column.

aws Search [Alt+S]

Amazon S3 > Buckets > results-ferran-aran

results-ferran-aran [Info](#)

[Objects](#) Metadata Properties Permissions Metrics Management Access Points

Objects (1)

[Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access yo

Find objects by prefix

☐	Name	Type	Last modified	Size
☐	project1/	Folder	-	

Inspecting the bucket from the AWS Console

And now we should see the file we just uploaded.

The screenshot shows the AWS S3 console interface. The browser address bar displays the URL: `us-east-1.console.aws.amazon.com/s3/buckets/results-ferran-aran?region=us-east-1&bucketType=general&prefix=project1/&showversions=false`. The console header includes the AWS logo, a search bar, and a notification bell. The breadcrumb navigation shows the path: `Amazon S3 > Buckets > results-ferran-aran > project1/`. The main content area is titled `project1/` and has two tabs: `Objects` (selected) and `Properties`. Below the tabs, there is a section titled `Objects (1)` with three action buttons: `Refresh`, `Copy S3 URI`, `Copy URL`, and `Download`. A descriptive text states: "Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to acc". Below this text is a search bar with the placeholder text `Find objects by prefix`. A table lists the objects in the bucket:

☐	Name	Type	Last modified
☐	new-file.txt	txt	March 4, 2025, 18:40:18 (UTC+01:00)

A red arrow points to the file name `new-file.txt` in the table.

Sync with a bucket

Syncing a local directory to download from a bucket

From the AWS CLI

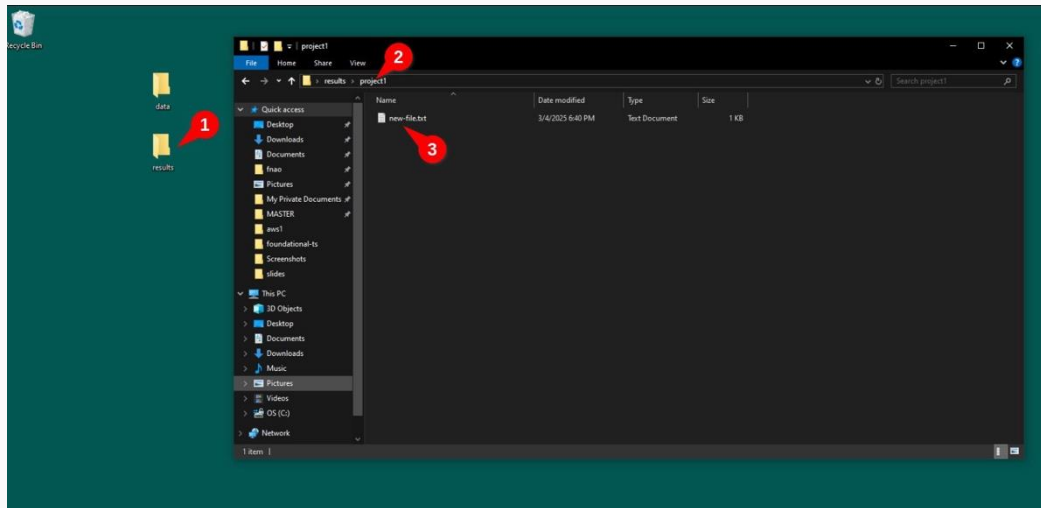
1. Create a new folder on your desktop and name it `results`.
2. Open a terminal and navigate to the folder. Use `cd Desktop`.
3. Run `aws s3 sync s3://results-{your-name} results` to download the files from the bucket.

In my case this is the result:

```
PS C:\Users\fnao\Desktop> aws s3 sync s3://results-ferran-aran results
download: s3://results-ferran-aran/project1/new-file.txt to results\project1\new-file.txt
PS C:\Users\fnao\Desktop>
```

Syncing a local directory to download from a bucket

We can use the File Explorer to navigate to the folder and see the file we just downloaded.



Recap

Recap - Today's contents

Today we have learned how to:

- Install the AWS CLI
- Configure AWS credentials
- Create an S3 bucket
- Sync a local directory to download from a bucket
- Sync a local directory to upload to a bucket
- Load files from the bucket to the notebook from python
- Write files from the notebook to the bucket from python

- With the setup we have done today we ended up with a bucket on which we can **upload datasets that we have to work on.**
- This datasets can be **accessed from any machine** with internet connection and the necessary permissions.
- We also have a results bucket which we can sync with our machine so **we have the results of our projects available locally.**

Recap - Subject's website

- Remember there is a website with useful information related to the subject.
- It has recently been updated with information about past sessions.
- Two new guides have been added to *Get started with AWS* and *Set up your Lab for every session*.